

The background features a dynamic, abstract composition. A thick, iridescent liquid stream flows from the top right, curving downwards and to the left. It is surrounded by numerous bubbles of varying sizes, some of which are also iridescent, while others are clear. The entire scene is set against a solid black background, creating a high-contrast, visually striking effect.

COFFEE SHOP SALES & CUSTOMER INSIGHTS ANALYSIS

Data-Driven Recommendations for Business Growth

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Executive Summary

This project analyzes sales and customer behavior data from a coffee shop to uncover key patterns and provide actionable recommendations. The analysis was conducted using Python (Jupyter Notebook) for data processing and visualization, and Power BI for interactive dashboard reporting. The study highlights trends in sales across time, identifies high-margin products, evaluates customer loyalty behavior, and examines seasonal preferences.

Key Findings:

- Morning hours drive the highest sales volume.
- Specialty coffee and bakery items are the most profitable combinations.
- Loyal customers spend significantly more than one-time buyers.
- Sales fluctuate by season, with iced drinks performing strongly in summer and hot beverages in winter.

Business Value:

By implementing targeted strategies around product promotion, customer loyalty, and operations, the coffee shop can increase revenue, improve customer satisfaction, and strengthen long-term growth.

Objectives & Methodology

Objectives

- Analyze sales trends across time periods and customer segments
- Identify high-performing and low-performing products.
- Understand customer loyalty and repeat purchase behavior.
- Provide data-driven business recommendations for growth.

Methodology

1. Data Collection & Cleaning – Imported sales and customer data, handled missing values, and standardized formats.
2. Exploratory Data Analysis – Used Python libraries to analyze sales patterns, customer behavior, and product performance.
3. Visualization – Created plots in Jupyter Notebook and dashboards in Power BI for interactive exploration.
4. Insights & Recommendations – Summarized findings and translated them into actionable strategies.

Tools Used

- Python: Pandas, NumPy, Matplotlib, Seaborn
- Jupyter Notebook: Data cleaning, analysis, and visualization
- Power BI: Interactive dashboards
- GitHub: Project documentation and version control



Dataset Overview

The dataset contains transaction-level details from a coffee shop, including product sales, customer data, and store information.

Key Features:

- Date & Time: Transaction timestamps (used for trend analysis).
- Product Category: Coffee, specialty drinks, bakery items, etc.
- Revenue: Total sales value per transaction.
- Customer ID: Used to measure repeat visits and loyalty.
- Store Location: Differentiates urban vs. suburban performance.

Data Preparation Steps:

- Removed duplicates and null values.
- Standardized product categories.
- Extracted time-based features (day, month, year, hour).
- Created customer segmentation based on purchase frequency.



An abstract, flowing liquid shape in vibrant colors (red, orange, yellow, green, blue, purple) swirling against a black background, resembling a stylized 'S' or a ribbon.

Insights

Sales Trends

- Peak sales occur in the morning (8–11 AM).
- Weekdays generate higher overall revenue than weekends.

Product Performance

- Coffee-based beverages dominate in sales volume.
- Specialty drinks like lattes and cappuccinos yield higher margins.
- Bakery items perform well as cross-sell products.

Customer Behavior

- Loyal customers generate higher repeat visits and spend more per order.
- Discounts increase traffic but reduce margins if overused.

Seasonality

- Iced drinks peak in summer months.
- Hot beverages dominate in colder months.
- Seasonal offerings (pumpkin spice, holiday specials) attract new buyers.

Store Performance

- Urban stores outperform suburban in volume.
- Suburban stores show higher average spend per customer.

Business Recommendations

Product Strategy

- Focus on promoting high-margin specialty drinks.
- Introduce seasonal limited-time offers to attract new customers.

Customer Loyalty

- Strengthen the loyalty program with tiered rewards.
- Use personalized offers based on purchase history.

Marketing Strategy

- Run targeted promotions during slower weekend periods.
- Use social media to promote seasonal drinks.

Operational Strategy

- Increase staff during morning rush hours to reduce wait times.
- Adjust inventory planning based on seasonal product demand.

Expansion Strategy

- Open new stores in urban high-traffic locations.
- Partner with delivery platforms in suburban areas to boost orders.



THANK YOU

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